

Lung and Colon Cancer Detection using EfficientNetB3

This project is developed to detect Lung and Colon Cancer from scanning images using the EfficientNetB3 deep learning model. The system allows users to upload CT scan or histopathology images and predicts whether the image belongs to lung cancer, colon cancer, or normal tissue. The project is implemented using Python, Flask, and TensorFlow/Keras.

Execution Process

- First, install Anaconda from the official website and complete the installation process. After installing Anaconda, open the Anaconda Prompt and create a new virtual environment using Python 3.11. Activate the created environment to install the required dependencies for the project.
- Next, install all the required libraries such as TensorFlow, Flask, NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, and Pillow. These libraries are necessary for image processing, model training, and building the web interface.
- After installing the required packages, download or clone the project files and open the project folder. Ensure that the dataset folder containing lung and colon cancer images is placed inside the project directory. The dataset should be organized into different folders based on cancer types and normal images.
- If the trained model file is not available, run the training script to train the EfficientNetB3 model using the dataset. The trained model will be saved in the project directory for prediction purposes.
- Once the model is ready, run the Flask application file using the Python command. The system will start running locally, and a local server link will be generated.
- Open the browser and navigate to the local server address. The project home page will be displayed with an option to upload scanning images. Upload the lung or colon scan image and click on the prediction button.
- The system preprocesses the image and passes it to the EfficientNetB3 model for prediction. The model analyzes the image and displays the predicted cancer type along with the result on the web interface.
- The project successfully detects lung and colon cancer from scanning images and provides fast and accurate predictions using the EfficientNetB3 deep learning model.

